

Experience

ML Research Engineer (Contract) — Ascentra, London 2025–

- Designed and delivered proof-of-concept for LLM integration into survey analysis product.
- Built custom chain-of-thought pipeline converting natural language queries to JSON requests.
- Delivered experiments comparing models for feasibility study and strategic recommendations.

Postgraduate Researcher — University College London 2024–

- Research on Neural Program Synthesis and Algorithmic Reasoning in the Machine Intelligence Lab, supervised by Prof. Mirco Musolesi. Developed novel deep learning framework.

Research Intern — Imperial College London 2023

- Causal discovery using cross-validation as compared with Bayesian Model Selection (using GPLVM), supervised by Mark van der Wilk and funded by Imperial UROP. HPC experience.

Education

University College London September 2023 – September 2024

MSc Machine Learning (*Distinction*)

MSc Thesis (*Distinction*) - supervised by Prof. Mirco Musolesi.

Hierarchical Bayesian Program Synthesis for Neural Algorithmic Reasoning

Probabilistic and Unsupervised Learning¹ | Approximate Inference¹ | Bayesian Deep Learning

Applied ML | Supervised Learning | Reinforcement Learning² | Applied Deep Learning

Imperial College London October 2020 – June 2023

BSc (Hons) Physics with Theoretical Physics (*2:1*)

BSc Thesis - supervised by Dr David Clements. *Computational Evolution*

Projects

Transformer Program Synthesis — University College London 2025

Developed novel programming-by-example framework: Transformer-based model recovers algorithms from data, training neural functions on-the-fly.

Masked Image Modelling — University College London 2024

Self-supervised learning framework: VisionTransformer pre-trained to reconstruct partially masked images and fine-tuned on Oxford-IIIT Pets dataset for segmentation. (*Distinction*)

Machine learning for particle physics — Imperial College London 2022

Lead ML team analysing decay product distribution from a Large Hadron Collider experiment (LHCb) for Standard Model comparison. TensorFlow, XGBoost. (*First-class, highest score*)

Solar Sail (Project Svarog) — Imperial College Space Society 2022

Orbital trajectory simulation, atmospheric effects, relativistic corrections. Launching in 2028.

Skills & Languages

Python (PyTorch, TensorFlow, GPflow, JAX), Julia; GPU clusters, distributed training; Git, Linux
English, Hungarian, French (Intermediate)

¹Module part of PhD programme at the Gatsby Computational Neuroscience Unit

²Module taught by Google DeepMind